National Testing Agency

Question Paper Name :	B Tech 2nd Apr 2026 Shift 1
Subject Name :	B. Tech
Creation Date :	2026-04-02 14:02:41
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Display Marks:	Yes

B. Tech

Group Number :	1
Group Id :	6911211
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Id :	6911211
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911211
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Id : 6911211 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\alpha, \alpha+2, \alpha \in \mathbf{Z}$, be the roots of the quadratic equation
$x(x+2)+(x+1)(x+3)+(x+2)(x+4)+\ldots.+(x+n-1)(x+n+1)=4n$ for some $n \in \mathbf{N}$.
Then $n+\alpha$ is equal to :

Options :

6911211. 0

6911212. 1

6911213. 2

6911214. 3

Question Number : 2 Question Id : 6911212 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let x and y be real numbers such that $50\left(\frac{2x}{1+3i}-\frac{y}{1-2i}\right)=31+17i$, $i=\sqrt{-1}$. Then the value of $10(x-3y)$ is :

Options :

6911215. 20

6911216. 31

6911217. 35

6911218. 75

Question Number : 3 Question Id : 6911213 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\alpha, \beta \in \mathbf{R}$ be such that the system of linear equations

$$x+2y+z=5$$
$$2x+y+\alpha z=5$$
$$8x+4y+\beta z=18$$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

Options :

6911219. -4

69112110. 4

69112111. 8

69112112. -8

Question Number : 4 Question Id : 6911214 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A=\begin{bmatrix}1 & 2\\ 1 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}3 & 3\\ \beta & 2\end{bmatrix}$. If $A^2-4A+I=O$ and $B^2-5B-6I=O$, then among the two statements :

(S1): $[(B-A)(B+A)]^T=\begin{bmatrix}13 & 15\\ 7 & 10\end{bmatrix}$

and

(S2): $\det(\text{adj}(A+B))=-5$,

Options :

69112113. only (S1) is correct

69112114. only (S2) is correct

69112115. both (S1) and (S2) are correct

69112116. both (S1) and (S2) are wrong

Question Number : 5 Question Id : 6911215 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let A be the set of first 101 terms of an A.P., whose first term is 1 and the common difference is 5 and let B be the set of first 71 terms of an A.P., whose first term is 9 and the common difference is 7. Then the number of elements in $A \cap B$, which are divisible by 3, is :

Options :

69112117. 4

69112118. 5

69112119. 6

69112120. 7

Question Number : 6 Question Id : 6911216 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of seven-digit numbers, that can be formed by using the digits 1, 2, 3, 5 and 7 such that each digit is used at least once, is :

Options :

69112121. 15400

69112122. 17800

69112123. 16800

69112124. 29400

Question Number : 7 Question Id : 6911217 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of elements in the set $S = \left\{(r, k) : k \in \mathbf{Z} \text{ and } {}^{36}C_{r+1} = \frac{6\left({}^{35}C_r\right)}{\left(k^2-3\right)}\right\}$, is :

Options :

69112125. 2

69112126. 4

69112127. 8

69112128. 16

Question Number : 8 Question Id : 6911218 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the mean of the data

Class	5-10	10-15	15-20	20-25	25-30	30-35
Frequency	2	k	28	54	k+1	5

is 21, then k is one of the roots of the equation :

Options :

69112129. $2x^2-23x-10=0$

69112130. $4x^2-35x+24=0$

69112131. $2x^2-19x-10=0$

69112132. $2x^2-35x+98=0$

Question Number : 9 Question Id : 6911219 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the mid points of the sides of a triangle ABC be $\left(\frac{5}{2}, 7\right)$, $\left(\frac{5}{2}, 3\right)$ and (4, 5). If its incentre is (h, k), then $3h + k$ is equal to :

Options :

69112133. 11

69112134. 12

69112135. 13

69112136. 14

Question Number : 10 Question Id : 69112110 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, a < b, pass through the point (4, 3) and have eccentricity $\frac{\sqrt{5}}{3}$. Then the length of its latus rectum is :

Options :

69112137. $\frac{4\sqrt{5}}{3}$

69112138. $2\sqrt{5}$

69112139. $\frac{7\sqrt{5}}{3}$

69112140. $\frac{8\sqrt{5}}{3}$

Question Number : 11 Question Id : 69112111 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\sin\left(\frac{\pi}{18}\right)\sin\left(\frac{5\pi}{18}\right)\sin\left(\frac{7\pi}{18}\right) = K$, then the value of $\sin\left(\frac{10K\pi}{3}\right)$ is :

Options :

69112141. $\frac{\sqrt{3}+1}{2\sqrt{2}}$

69112142. $\frac{\sqrt{3}-1}{\sqrt{2}}$

69112143. $\frac{\sqrt{3}}{2}$

69112144. $\frac{1}{2}$

Question Number : 12 Question Id : 69112112 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $S=\{x\in[-\pi,\pi] : \sin x\,(\sin x+\cos x)=a,\ a\in Z\}$. Then n(S) is equal to :

Options :

69112145. 3

69112146. 6

69112147. 7

69112148. 9

Question Number : 13 Question Id : 69112113 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the point of intersection of the lines $\frac{x+1}{3} = \frac{y+a}{5} = \frac{z+b+1}{7}$ and $\frac{x-2}{1} = \frac{y-b}{4} = \frac{z-2a}{7}$ lies on xy-plane, then the value of $a+b$ is :

Options :

69112149. 2

69112150. 5

69112151. 7

69112152. 9

Question Number : 14 Question Id : 69112114 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 2$ and $|\vec{b}| = 3$, then the maximum value of $3\left|\left(3\vec{a} + 2\vec{b}\right)\right| + 4\left|\left(3\vec{a} - 2\vec{b}\right)\right|$ is :

Options :

69112153. 30

69112154. 36

69112155. 60

69112156. 72

Question Number : 15 Question Id : 69112115 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let a line L passing through the point (1, 1, 1) be perpendicular to both the vectors $2\hat{i} + 2\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 2\hat{k}$. If P(a, b, c) is the foot of perpendicular from the origin on the line L, then the value of $34(a+b+c)$ is :

Options :

69112157. 50

69112158. 80

69112159. 100

69112160. 120

Question Number : 16 Question Id : 69112116 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\lim\limits_{x\to 2} \dfrac{\sin\left(x^3-5x^2+ax+b\right)}{\left(\sqrt{x-1}-1\right)\log_e(x-1)} = m$, then $a+b+m$ is equal to :

Options :

69112161. 5

69112162. 6

69112163. 8

69112164. 10

Question Number : 17 Question Id : 69112117 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the curve $y=f(x)$ passes through the point $(1, e)$ and satisfies the differential equation $dy=y(2+\log_e x)\,dx$, $x>0$, then $f(e)$ is equal to :

Options :

69112165. e^e

69112166. e^{e^2}

69112167. e^{2e}

69112168. e^{2^e}

Question Number : 18 Question Id : 69112118 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of critical points of the function $f(x)=\begin{cases}\left|\dfrac{\sin x}{x}\right| & , x\neq 0\\ 1 & , x=0\end{cases}$ in the interval $(-2\pi, 2\pi)$ is equal to :

Options :

69112169. 1

69112170. 3

69112171. 5

69112172. 7

Question Number : 19 Question Id : 69112119 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $[\cdot]$ denote the greatest integer function. Then the value of $\int_0^3 \left(\frac{e^x+e^{-x}}{[x]!}\right) dx$ is :

Options :

69112173. $e^2 + e^3 - \frac{1}{e^2} - \frac{1}{e^3}$

69112174. $\frac{1}{2}\left(e^2 + e^3 - \frac{1}{e^2} - \frac{1}{e^3}\right)$

69112175. $e^2 + e^3 - \frac{1}{2e^2} - \frac{1}{2e^3}$

69112176. $\frac{1}{2}(e^2 + e^3) - \frac{1}{e^2} - \frac{1}{e^3}$

Question Number : 20 Question Id : 69112120 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $y = y(x)$ be the solution curve of the differential equation $(1 + \sin x)\frac{dy}{dx} + (y+1)\cos x = 0, y(0) = 0$. If the curve $y = y(x)$ passes through the point $\left(\alpha, \frac{-1}{2}\right)$, then a value of α is :

Options :

69112177. $\frac{\pi}{6}$

69112178. $\frac{\pi}{4}$

69112179. $\frac{\pi}{3}$

69112180. $\frac{\pi}{2}$

Mathematics Section B

Section Id :	6911212
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory

Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911212
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Id : 69112121 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the domain of the function $f(x)=\sqrt{\log_{(0.6)}\left(\left|\frac{2x-5}{x^2-4}\right|\right)}$ is $(-\infty, a] \cup \{b\} \cup [c, d) \cup (e, \infty)$, then the value of $a+b+c+d+e$ is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 22 Question Id : 69112122 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $\sum_{k=1}^{n} a_k = 6n^3$, then $\sum_{k=1}^{6}\left(\frac{a_{k+1}-a_k}{36}\right)^2$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 23 Question Id : 69112123 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $a, b, c \in \{1, 2, 3, 4\}$. If the probability, that $ax^2+2\sqrt{2}bx+c>0$ for all $x \in \mathbf{R}$, is $\frac{m}{n}$, $\gcd(m, n)=1$, then $m+n$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Question Number : 24 Question Id : 69112124 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let a circle C have its centre in the first quadrant, intersect the coordinate axes at exactly three points and cut off equal intercepts from the coordinate axes. If the length of the chord of C on the line $x+y=1$ is $\sqrt{14}$, then the square of the radius of C is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
1

Question Number : 25 Question Id : 69112125 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $\alpha = \int_0^{2\sqrt{3}} \log_2\left(x^2+4\right)dx + \int_2^4 \sqrt{2^x-4}\,dx$, then α^2 is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

Physics Section A

Section Id :	6911213
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	6911213
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 69112126 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The dimensional formula of $\frac{1}{2}\epsilon_o E^2$ (ϵ_o = permittivity of vacuum and E = electric field) is $M^aL^bT^c$.

The value of $2a-b+c=$_____.

Options :

69112186. 0

69112187. 1

69112188. -1

69112189. 2

Question Number : 27 Question Id : 69112127 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical